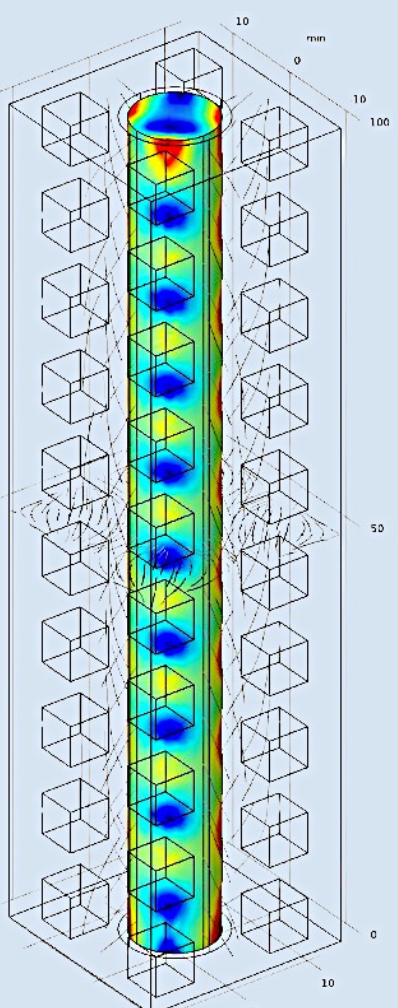
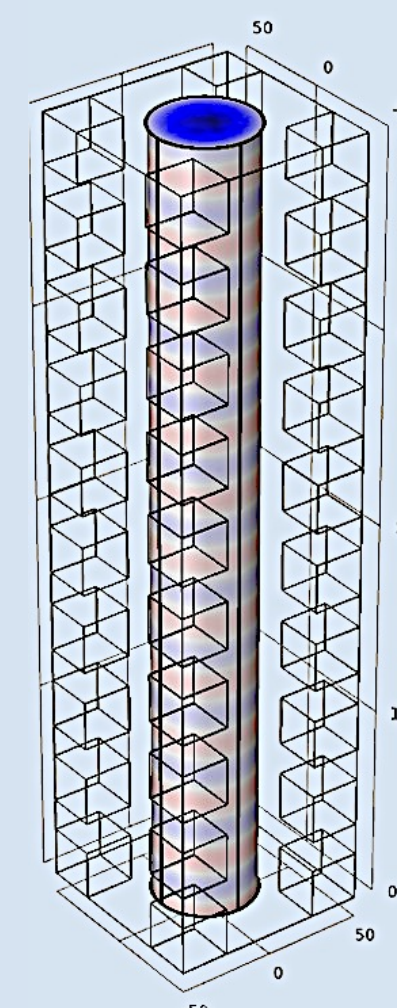




Graphic created by Finalist using COMSOL Multiphysics® 5.6, 2025

Anderson Localization in a Ferrofluidic Waveguide System with a Topological Defect



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INTRODUCTION

Anderson Localization

- Absence of wave diffusion in random lattices
- Wave **isolation** in **disordered crystal**

Success in Photonics

- Realization of Anderson Localization via:
 - Superlattice structures
 - Optical Fibers
 - Semiconductor Powders

Impracticality in Phononics

- Failed methods:
 - Suspended air bubbles in fluid medium
 - Random-sized spherical scatterers
 - Randomly-located beam scatterers

Inducing Disorder in Crystals

- Eliminate Characteristic Brillouin Zone
 - Manipulate Fluid-Fluid Phononic Crystal with solid-like nanoparticles
- Topological Defect
 - Soliton propagation through point aperiodicity

PURPOSE

- Design an **experimentally-feasible Ultrasound Isolation Chamber** to physically realize Anderson Localization

APPROACH

- Subject **Ferrofluid** inside **Waveguide** to controlled magnetic field via **standard magnets**
 - Overall: Fluid-Fluid Phononic Crystal
 - Nanoparticles: solid-like behavior with individual reactions to the magnetic field
- Induce **Topological Defect** to enhance potency of Anderson Localization effects
- Simulate** Design via COMSOL Multiphysics® 5.6 Software
 - Finite Element Method (FEM) Scheme for Magnetic Flux Density
 - Finite-Difference Time-Domain (FDTD) Scheme for Acoustic Pressure

METHODOLOGY

Literature Review

- Analyze attempted methods at realizing Anderson Localization
- Determine ideal Ferrofluid properties for simulation
- Record sinusoidal curve-fitting algorithms used

Equipment Review

- Observe physical-testing requirements for waveguides
- Tailor model for practical use in experiments

COMSOL Multiphysics® CAD Kernel

- Design preliminary waveguide with solenoids and magnets
- Redesign and simplify
- Control Group and Two Topological Defect Structures

COMSOL Multiphysics® Multiphysics Kernel

- Set Metamaterial and standard magnet Magnetization properties
- Propagate Acoustic wave through cutline traversing waveguide
- Generate "Extremely Fine" data set (2500 data points per model)

Analysis

- Convert data points to 2D-graph traversing arc length
- Apply sinusoidal curve-fitting algorithms to compare models
- Empirically determine presence of Anderson Localization effects

Chart created by Finalist, 2025

MAGNET SCHEME

Magnet Set	Magnet Center of Mass Height	Gap Center of Mass Height
1	12.5	33.3
2	54.16	75
3	95.83	166.6
4	137.5	158.3 + $\frac{p}{2}$
5	179.16 + p	200
6	220.83 - p	214.6 - $\frac{p}{2}$
7	262.5	283.3
8	304.16	325
9	345.83	366.6
10	387.5	

Data Table created by Finalist, 2025

Parameter p adjusts the locations of the fifth and sixth sets of **N52 Magnets** to create the **Topological Defect**.

NOTATION

Magnetic Field Theory

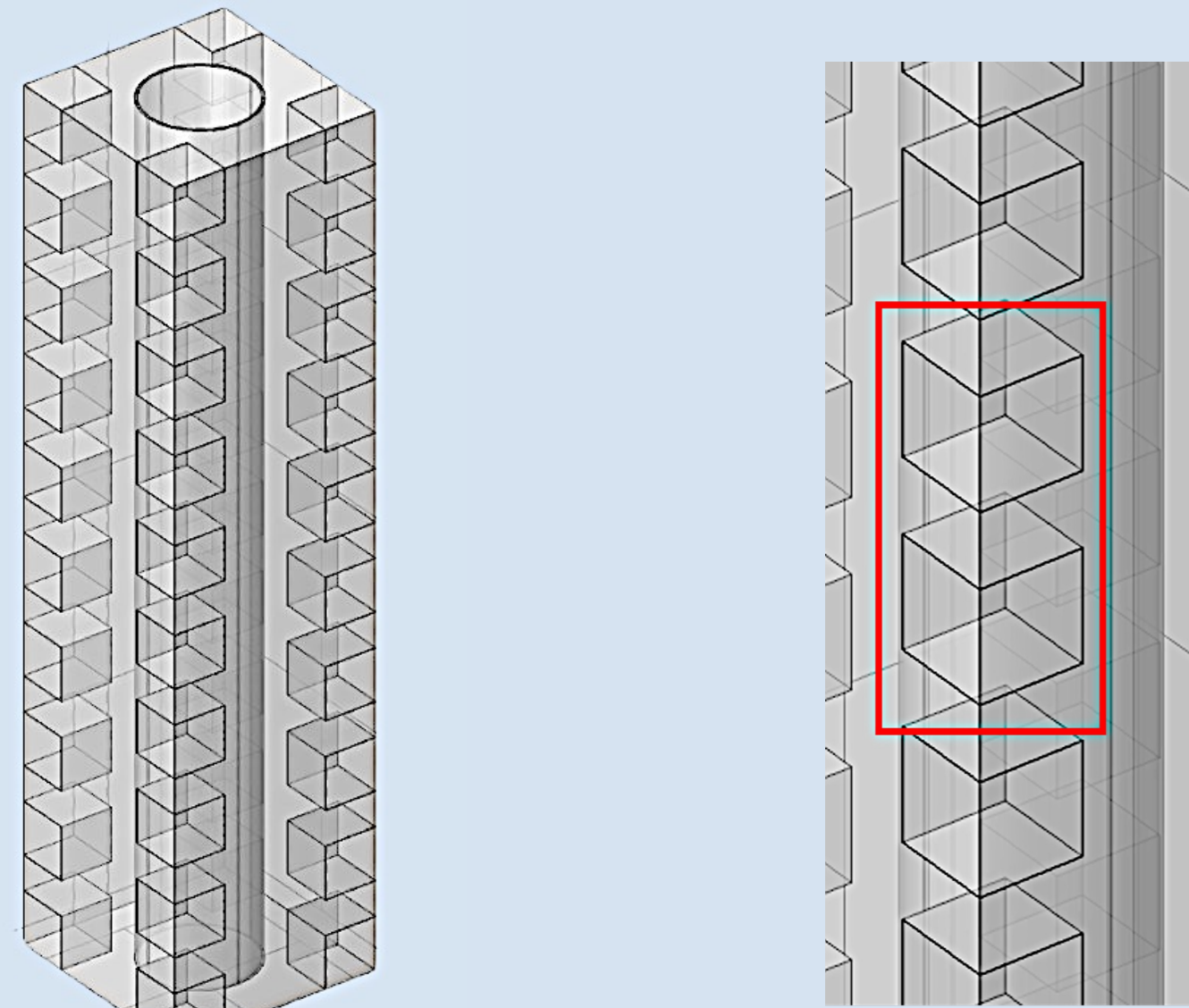
- $\vec{H}$: Magnetic Field Strength [A/m]
- V_m : Magnetic Scalar Potential [A]
- $\vec{B}$: Magnetic Flux Density [T]
- μ_0 : Vacuum Magnetic Permeability, $[4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}]$
- $\vec{M}$: Magnetization [A/m]
- $\vec{D}$: Electric Displacement Field
- ϵ_0 : Vacuum Permittivity Constant, $[8.85 \times 10^{-12} \text{ C}^2/(\text{N} \cdot \text{m}^2)]$
- $\vec{E}$: Electric Field [N/C]
- $\vec{P}$: Polarization Density [C/m²]
- $\vec{J}$: Current Density [A/m²]
- $\sigma \equiv \frac{1}{\rho}$: Conductivity, inverse of resistivity $[1/(\Omega \cdot \text{m})]$
- μ_r : Relative Permeability
- $\vec{B}_r$: Remnant Magnetic Flux Density [T]
- $\vec{n}$: Boundary Normal Vector
- $\vec{A}$: Area Vector [m²]
- m : Azimuthal Mode Number
- ω : Angular Frequency [rad/s]
- L_p : Sound Pressure Level [dB]
- p : Acoustic Pressure [Pa]
- ρ : Medium Density [kg/m³]
- q_d : Dipole Domain Source [N/m³]
- Q_m : Monopole Domain Source [1/s²]
- $k_m \equiv \frac{m}{r}$: Azimuthal Wave Number [1/m]
- c_c : Speed of Sound in Medium [m/s]

Acoustic Pressure Theory

- m : Azimuthal Mode Number
- ω : Angular Frequency [rad/s]
- L_p : Sound Pressure Level [dB]
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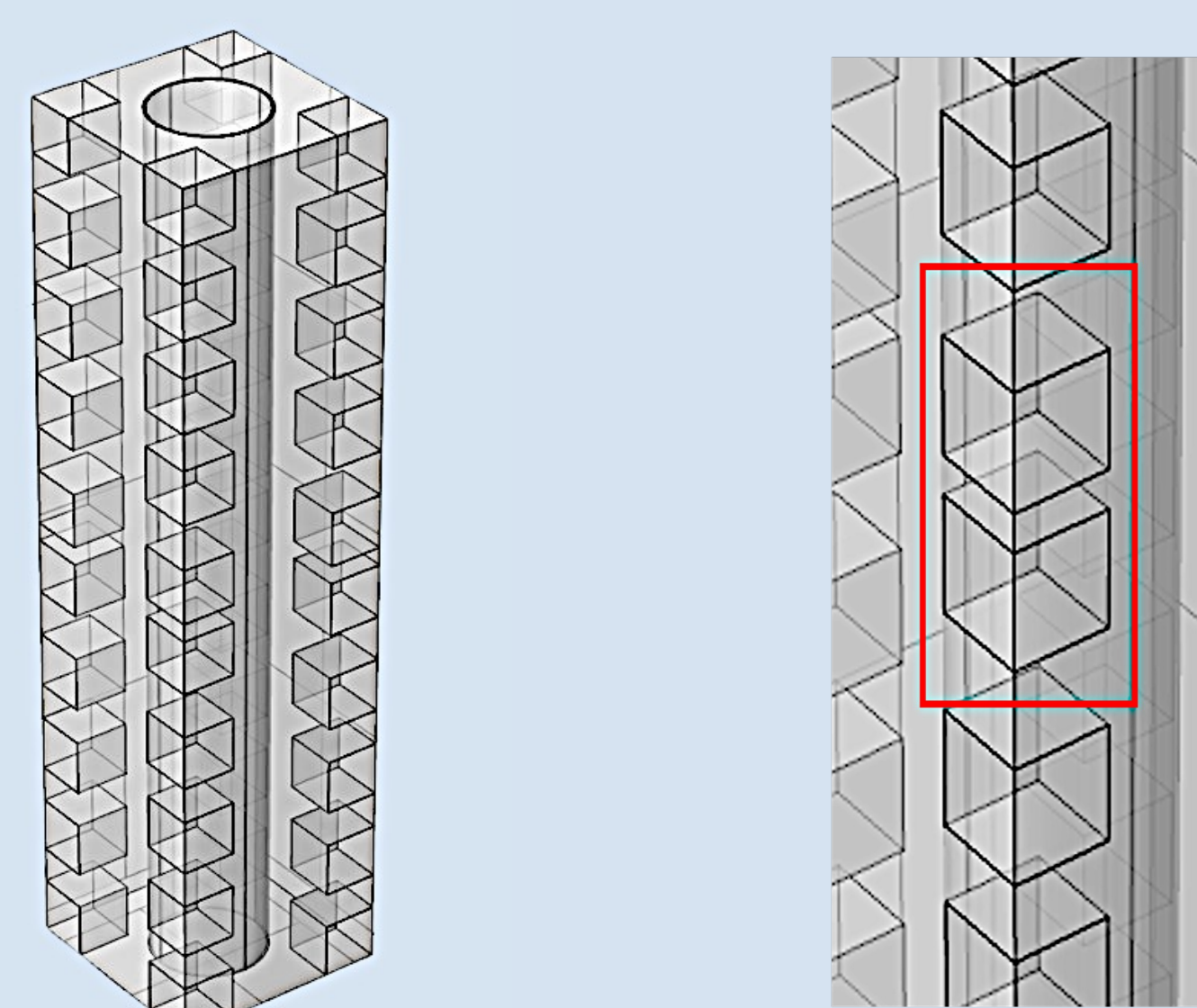
MODELS

Periodic Structure: $p = 0$



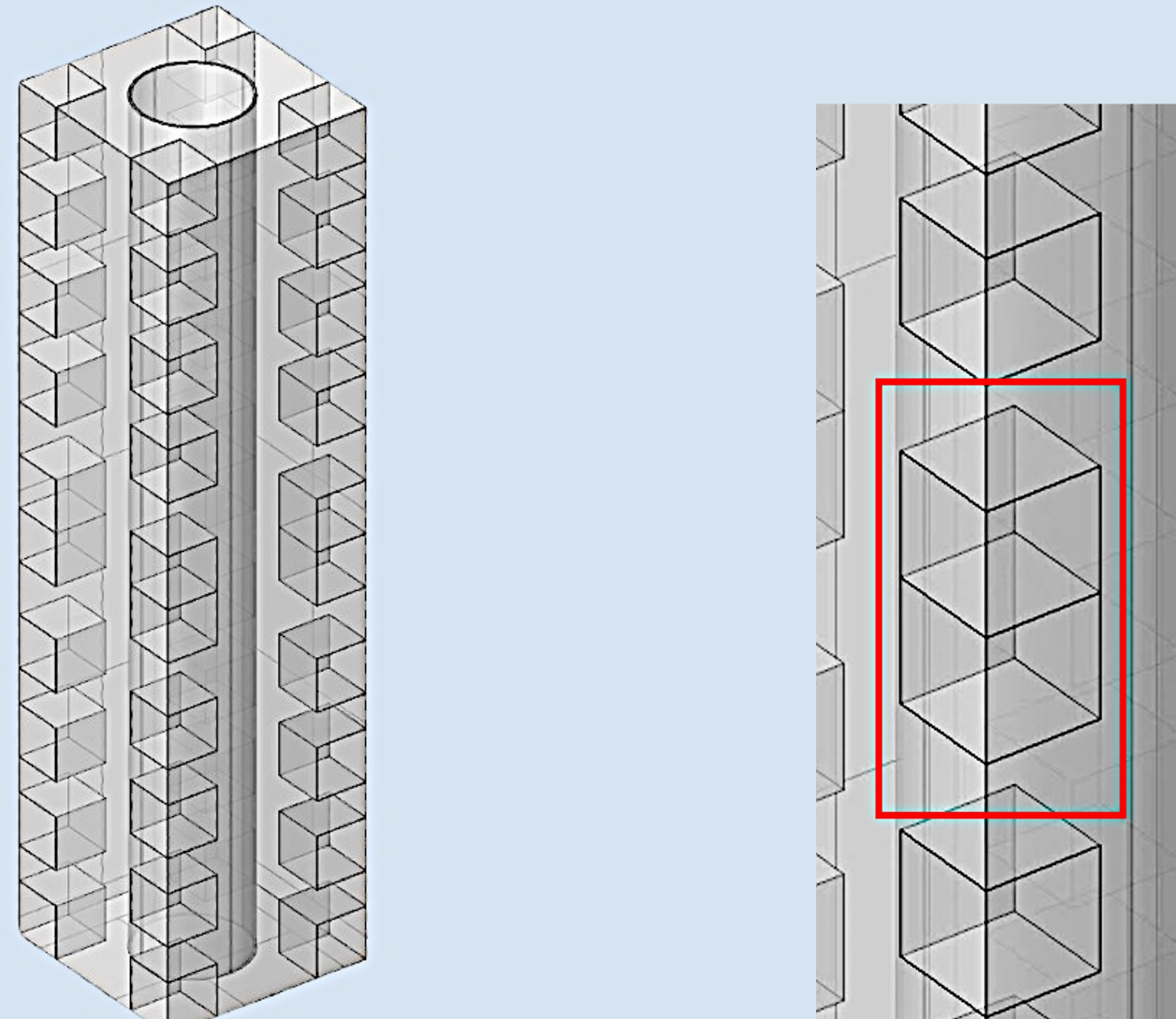
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Half-Defect Structure: $p = 25/6$



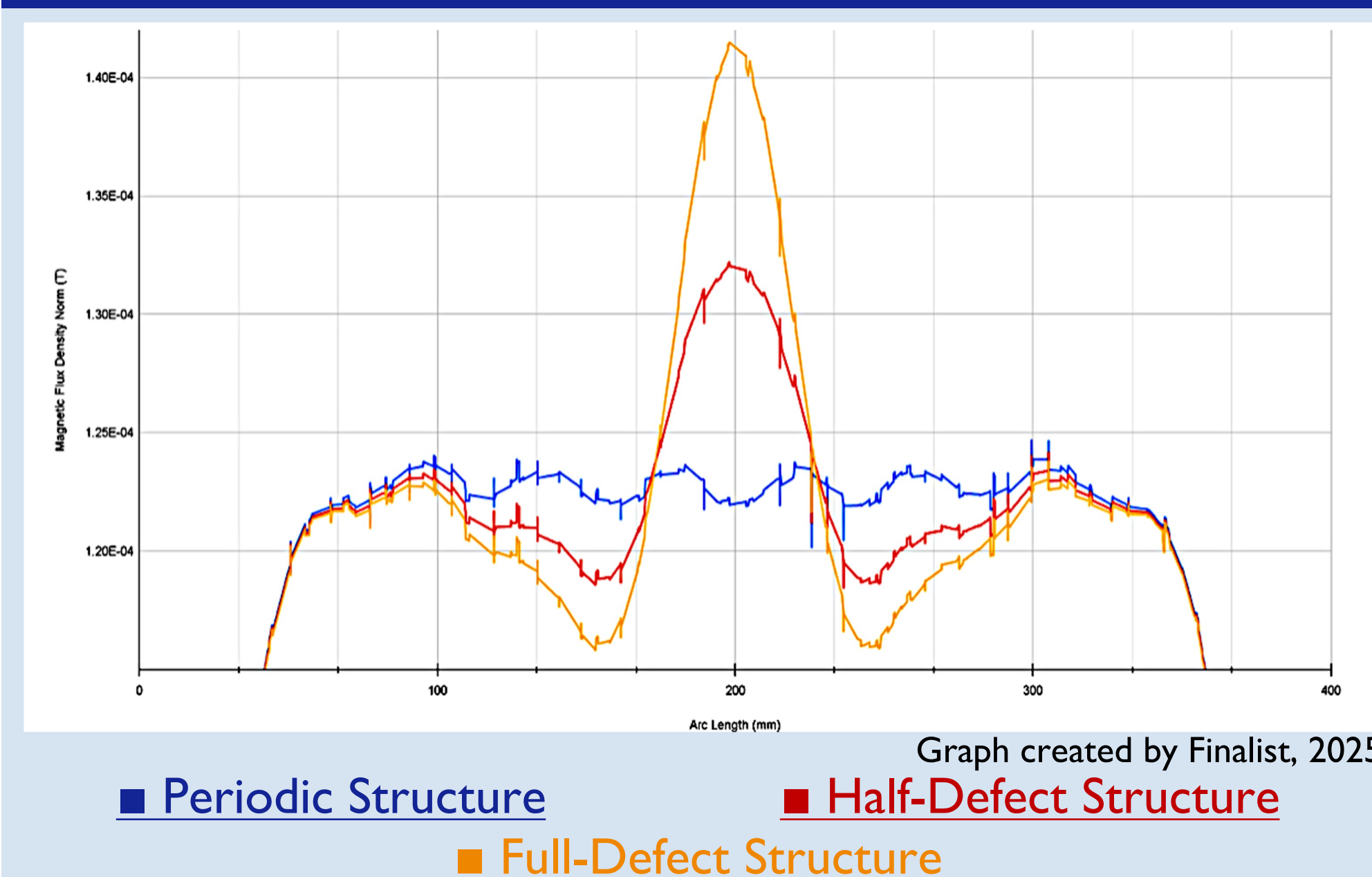
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Full-Defect Structure: $p = 25/3$



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MAGNETIC FLUX DENSITY



MAGNETIC FIELD THEORY

Magnetic Fields, No Currents

- Current-free region:
$$\nabla \times \vec{H} = 0 \Rightarrow \vec{H} \equiv -\nabla V_m$$
- $\vec{B} = \mu_0(\vec{H} + \vec{M})$ and $\nabla \cdot \vec{B} = 0$ [Maxwell]
$$\Rightarrow -\nabla \cdot (\mu_0 \nabla V_m - \mu_0 \vec{M}_0) = 0$$

Magnetic Flux Conservation

- Constitutive Relations:
 - $\vec{D} \equiv \epsilon_0 \vec{E} + \vec{P}$ [$\vec{D}$ Field Definition]
 - $\vec{B} = \mu_0(\vec{H} + \vec{M})$ [$\vec{H}$ Field Definition]
 - $\vec{j} = \sigma \vec{E}$ [Ohm]
- Generalized Constitutional Relation B-H:
$$\vec{B} = f(\|\vec{H}\|) = \mu_0 \mu_r \vec{H} + \vec{B}_r$$

- Simulate with Material B-H (Demagnetization) Curve

Magnetic Insulation

- Boundary Condition on tangential component of Magnetic Potential: $\vec{n} \times \vec{A} = 0$

Magnetic Flux Density

- Boundary Condition on normal component of Magnetic Flux Density: $\vec{n} \cdot \vec{B} = \vec{n} \cdot \vec{B}_0$

ACOUSTIC PRESSURE THEORY

Pressure Acoustics

- 2-dimensional axisymmetric cross-section:
$$p(r, \varphi, z) = p(r, z)e^{-im\varphi}$$
- Second Order Nonreflecting Boundary Condition Approximation
- Default Scaling: $\Delta = \frac{1}{\omega^2}$
- $L_p = 20 \log\left(\frac{p_{\text{RMS}}}{p_{\text{ref}}}\right)$ for $p_{\text{RMS}} \equiv \sqrt{\frac{1}{2}pp^*}$
- Default Discretization: Quadratic Lagrange

Frequency (Time-Harmonic) Domain

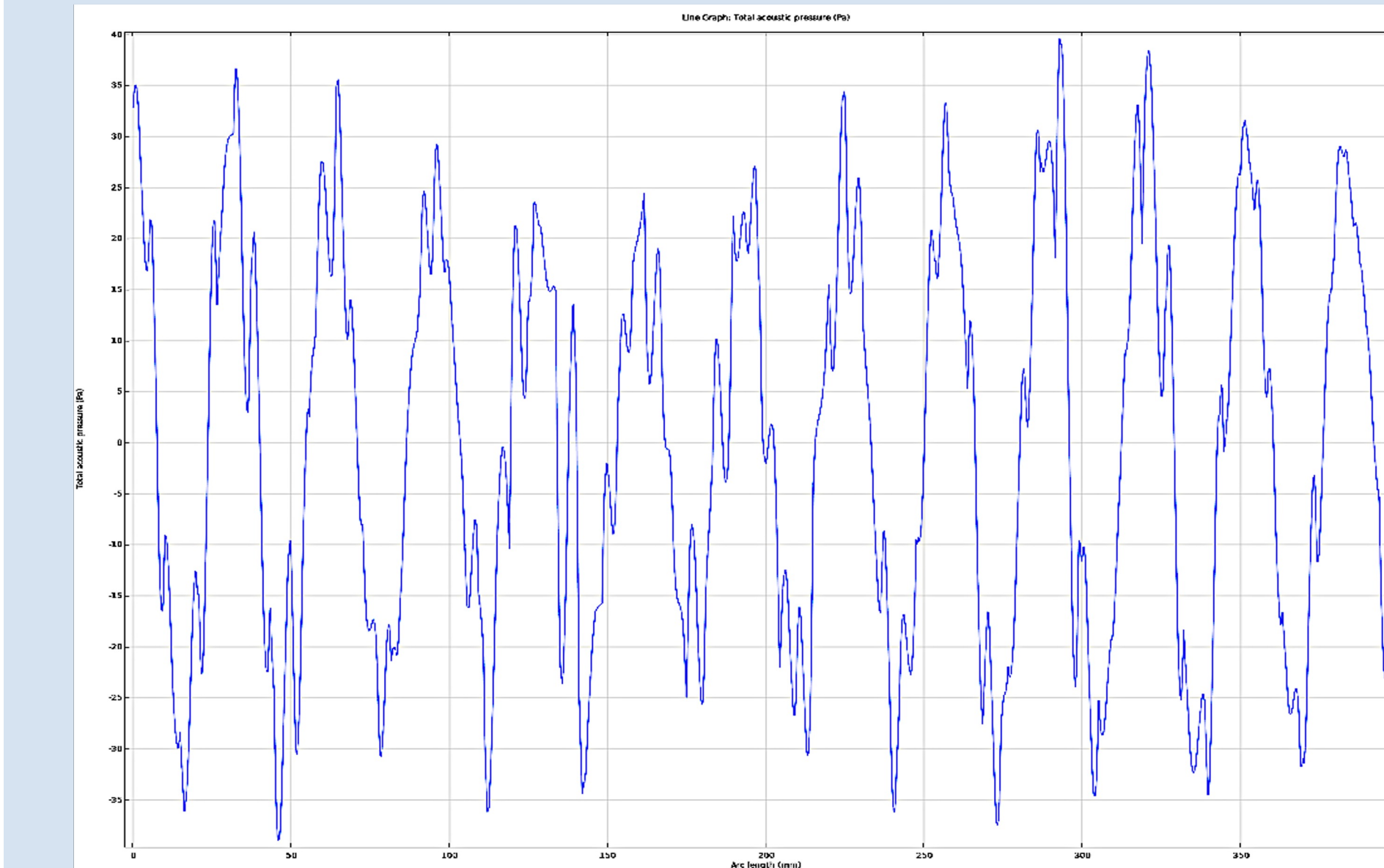
- Inhomogeneous Helmholtz Equation:
$$\nabla \cdot \left(-\frac{1}{\rho_c} (\nabla p_{\text{total}} - q_d) \right) - \frac{k_{\text{eq}}^2 p_{\text{total}}}{\rho_c} = Q_m$$
$$\Rightarrow \frac{\partial}{\partial r} \left[-\frac{r}{\rho_c} \left(\frac{\partial p}{\partial r} - q_r \right) \right] + r \frac{\partial}{\partial z} \left[-\frac{1}{\rho_c} \left(\frac{\partial p}{\partial z} - q_z \right) \right] - \frac{k_{\text{eq}}^2}{\rho_c} r p = r Q_m$$
$$\Rightarrow k_{\text{eq}}^2 = \left(\frac{\omega}{c_c} \right)^2 - k_m^2$$

Wave Propagation

- 500 kHz Trigger Peak propagated in z-direction to model standard transducer equipment

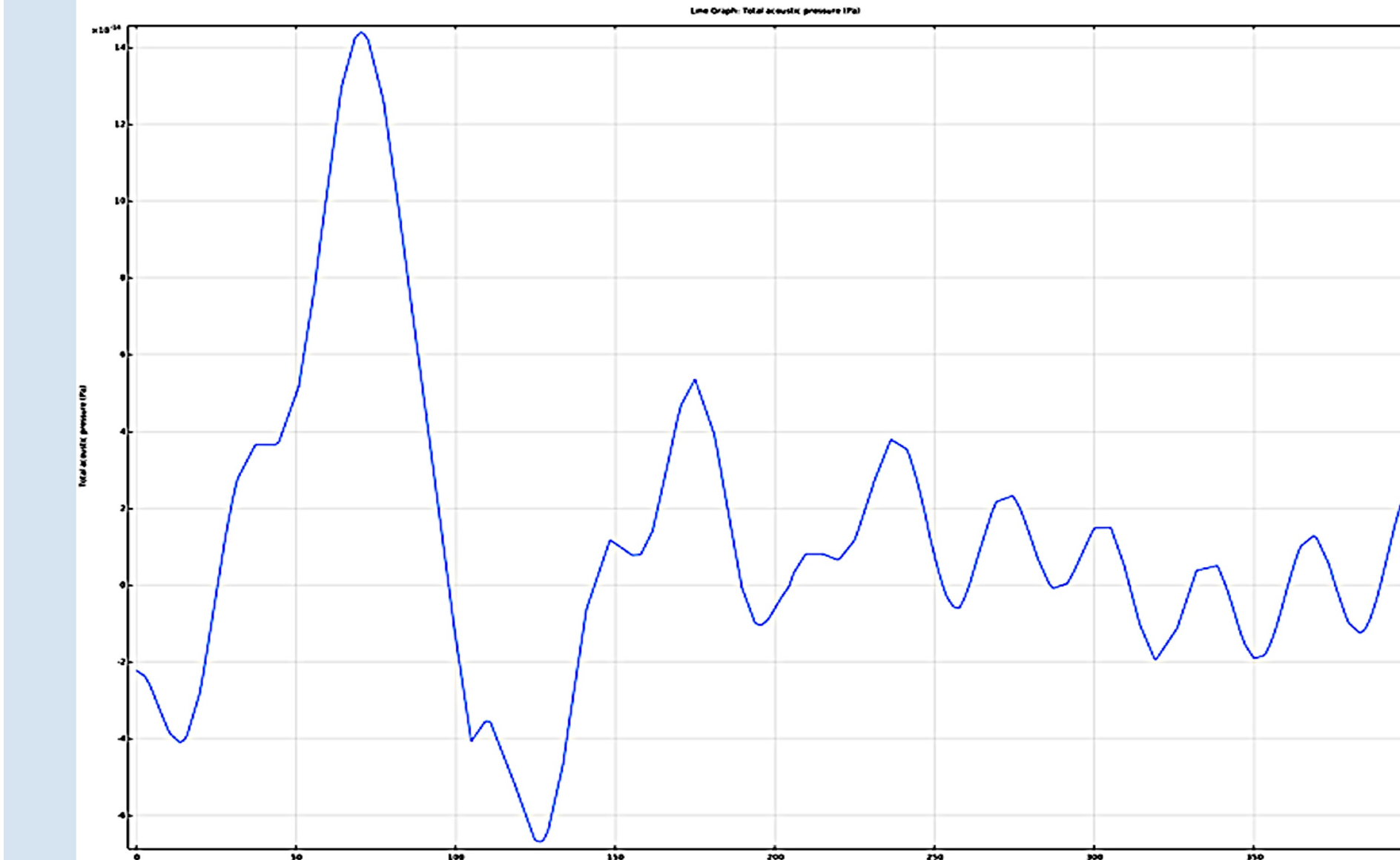
ACOUSTIC PRESSURE

Periodic Structure: $p = 0$



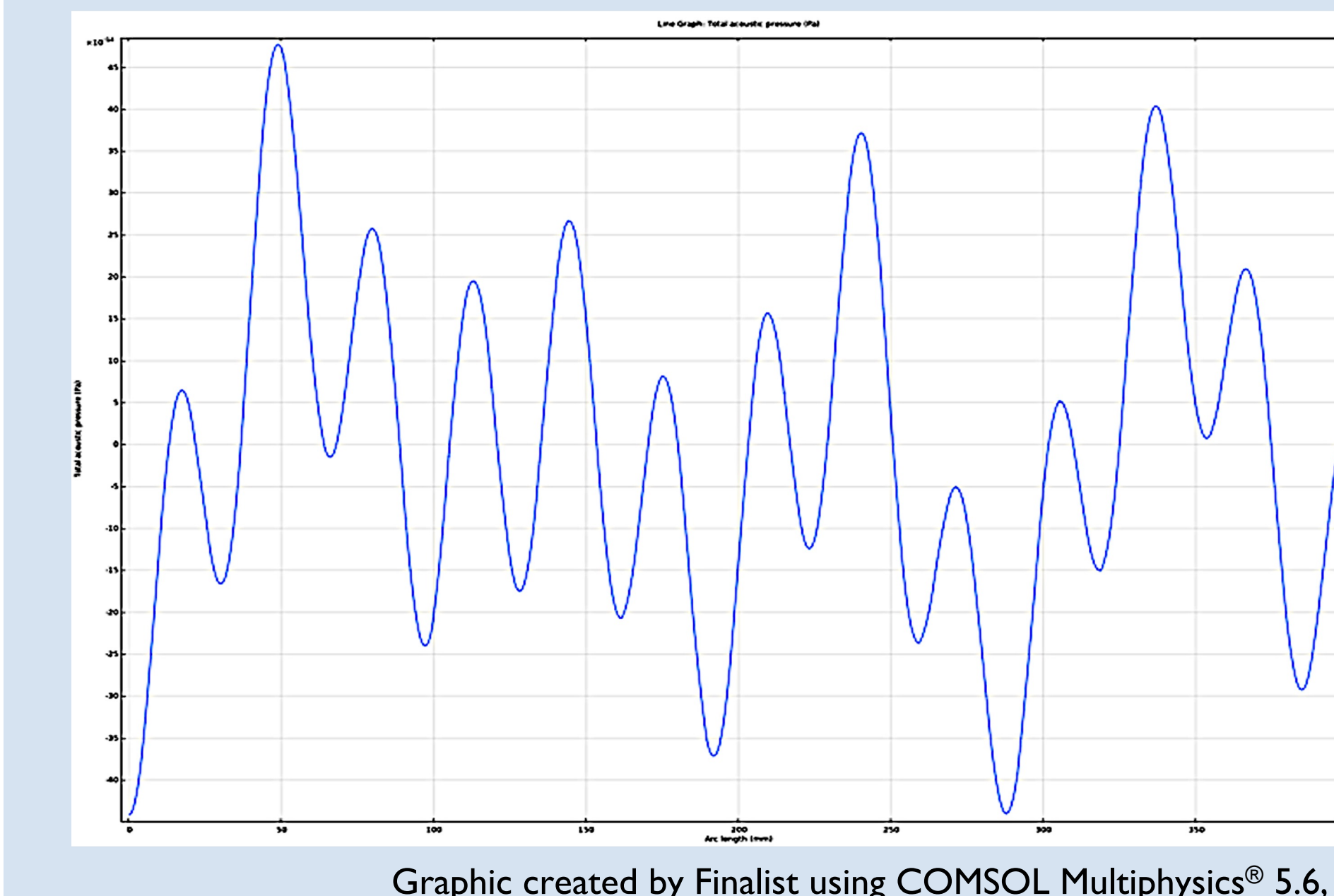
Graphic created by Finalist using COMSOL Multiphysics® 5.6, 2025

Half-Defect Structure: $p = 25/6$



Graphic created by Finalist using COMSOL Multiphysics® 5.6, 2025

Full-Defect Structure: $p = 25/3$



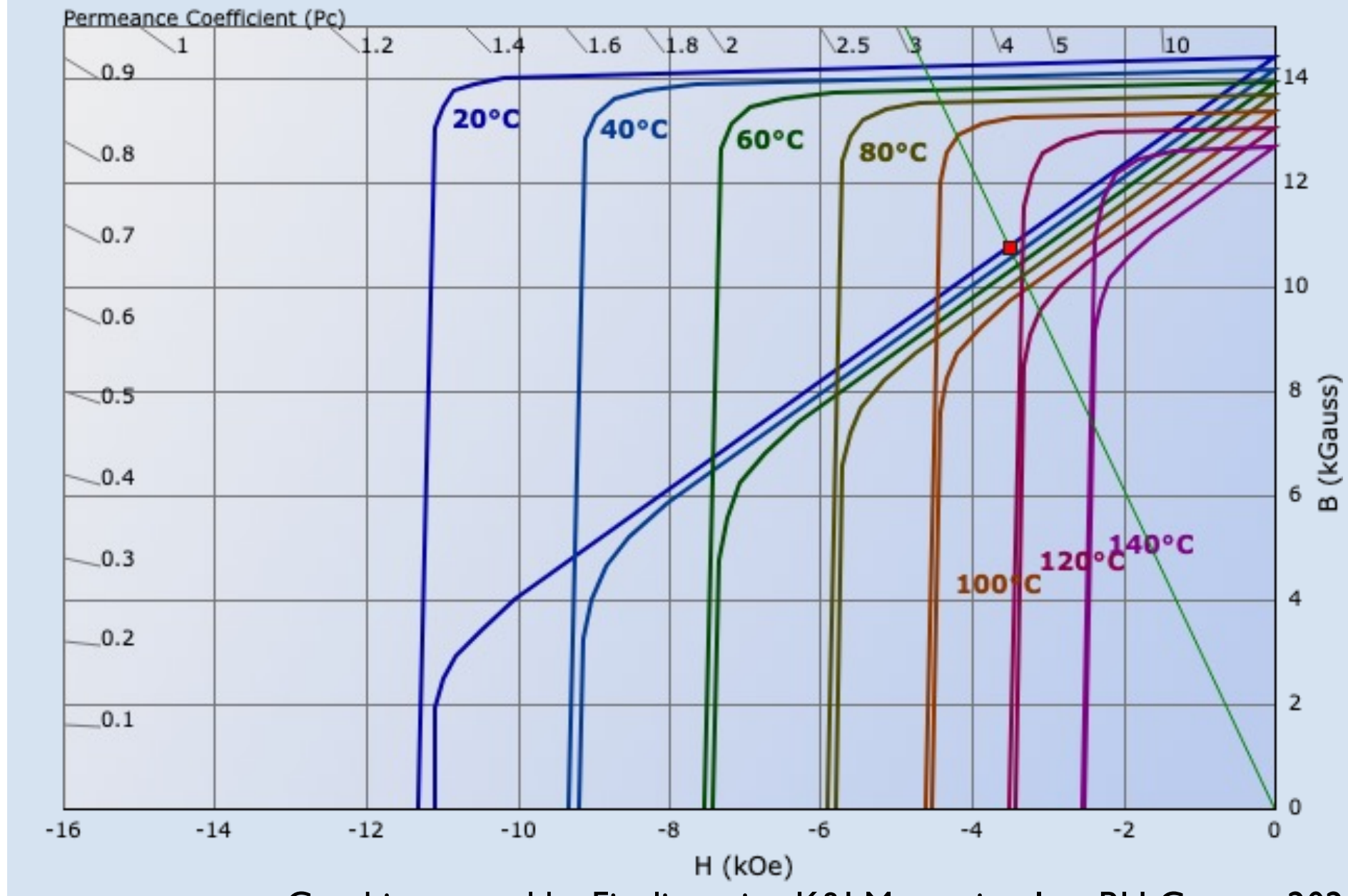
Graphic created by Finalist using COMSOL Multiphysics® 5.6, 2025

MATERIAL SPECIFICATIONS

N52 Magnet Specifications

- 25 mm × 25 mm × 25 mm N52-Grade Block-type
- Surface Field: $\vec{B} = 0.6451 \text{ T}$
- Permeance Coefficient: $P_c = 3.07$
- Dipole Moment: $\vec{m} = 18.4 \text{ A} \cdot \text{m}^2$
- Magnetization: $\vec{M} = \frac{\vec{m}}{V} = \frac{18.4 \text{ A} \cdot \text{m}^2}{(25 \times 10^{-3} \text{ m})^3} = 1177600 \frac{\text{A}}{\text{m}}$

N52 Magnet B-H (Demagnetization) Curve



Graphic created by Finalist using K&J Magnetics, Inc. BH Curves, 2025

Load Line and Operating Point based on Room Temperature of 20°C

Ferrofluid Metamaterial Specifications

- Ferrotec PBG900-model Polyethylene Glycol (PEG) Oil-based Ferrofluid
- Specialized for industrial usage in Sensors, Switches, and Solenoids
- Saturation Magnetization $\|\vec{M}_s\| = 99 \text{ mT}$
- Viscosity (27°C) $\mu < 40 \text{ mPa} \cdot \text{s}$
- Waveguide Specifications
 - 25 mm × 25 mm × 400 mm Circular Tubing type
 - Phononic Crystal Waveguide using Semiconducting materials; Silicon used in simulation
- Environment Specifications
 - Air environment to model physical experiments

DATA ANALYSIS

Control Group: Periodic Structure

- Expected behavior: periodic Magnetic Flux Density
- Expected behavior: **sinusoidal** Acoustic Pressure plot

Test Case: Half-Defect Structure

- Acoustic Pressure damping by **2 orders of magnitude**
- Aperiodic Spike in Magnetic Flux band structure
- Edge Case: Full-Defect Structure
 - Erratic Acoustic Pressure: excessive division of control peaks and multiple aperiodic points

ERROR ANALYSIS

- Computational Power Shortage
 - Decreased localization potency
- Potentially expensive for experimentation

RESULTS

Successful Topological Defect

- Half-Defect Structure exhibited a local minimum at the aperiodic point
- Combining and damping of control echoes
- Evidence of Anderson Localization
- Echoes of **smaller peak-to-peak amplitude** compared to the Trigger peak
- Acoustic Pressure **approaches 0 Pa** 400 mm away from the trigger peak

CONCLUSION

Potential for Greater Anderson Localization Effects

- Large limit of arc length: Acoustic Pressure will tend to 0 Pa
- Experimental Feasibility
 - All components readily available
 - Versatility in physical applications
- Efficiency** in achieving Anderson Localization in **simulations** and has potential for efficient **physical observation** of Anderson Localization

APPLICATIONS

Commercial Market: Media Entertainment Systems

- Cost Efficiency**
 - Ferrofluid implementation in hardware to achieve true "Noise Cancellation"
 - Reduced need for wave-inverting software
- Increased Sound Output** for consumers
 - Avoid accidental localization in majority of speakers made with Ferrofluid

All photos, images, and graphics were done by Finalist unless otherwise stated